

HumVis: Human-Centric Visual Analysis System

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ABSTRACT

Human-centric visual analysis is a fundamental task for many multimedia and computer vision applications, such as self-driving, multimedia retrieval, and augmented reality, etc. Based on our recent research efforts on fine-grained human visual analysis, we develop a robust and efficient human-centric visual analysis system named as HumVis. HumVis is built on a simple yet efficient contextual instance decoupling (CID) module, which can effectively separate different persons in an input image and output corresponding person structure information for visual analysis. Based on CID, HumVis achieves accurate multi-person pose estimation, multi-person foreground segmentation, multi-person part segmentation and 3D human mesh recovery for user-uploaded images/videos and support live stream presentation.

CCS CONCEPTS

• **Computing methodologies** → **Computer vision.**

KEYWORDS

Multi-Person Pose Estimation, Multi-Person Segmentation, Multi-Person Part Segmentation, 3D Human Mesh Recovery

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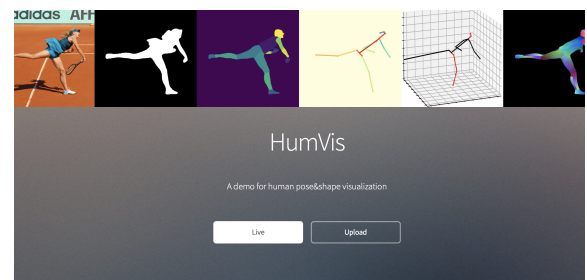


Figure 1: The main interface of HumVis.

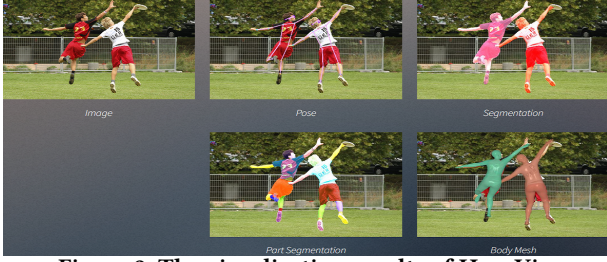
1 INTRODUCTION

The rapid progresses in Computer Vision and Deep Learning have pushed the human analysis research beyond category-level detection towards fine-grained instance-level understanding. Human-centric visual analysis aims to estimate human fine-grained pose and shape information from input RGB images, which plays a key role for many high-level multimedia and vision tasks, *e.g.*, action recognition [10], person re-identification [9], animation [11] and rendering [16]. It is also involved in widespread applications such as VR/AR content creation, virtual try-on, and dynamic simulations.

Crowded scene in the wild is one fundamental challenge of human-centric visual analysis, where multiple persons are overlapped with each other. Decoupling persons into different representations is an essential step in existing human-centric visual analysis methods. Existing ways of decoupling persons can be summarized into *i.e.*, top-down [6, 15, 18, 21], bottom-up [2, 7, 8, 12, 14, 17], and single-stage regression methods [4, 5, 13, 19, 22, 23], respectively. The above pipelines show different properties and defects. A detailed review of existing methods can refer to [20]. We propose a new pipeline named contextual instance decoupling (CID) [20] to effectively decouple instances from crowded scenes while preserving rich context cues for fine-grained instance-level human analysis. Given an image with multiple persons, we first decouple each person into a specific instance-aware feature map, which

Table 1: Comparison with recent works on several human-centric visual analysis tasks.

Method	Type	Pose Estimation			Foreground Segmentation			Body Part Segmentation			
		AP	AP ^M	AP ^L	AP	AP ^M	AP ^L	mIoU	AP ^P ₅₀	AP ^P _{vol}	PCP ₅₀
Mask R-CNN [6]	top-down	63.1	57.8	71.4	53.2	43.3	64.8	47.2	41.4	45.4	44.3
PersonLab [1]	bottom-up	66.5	62.4	72.3	-	47.6	59.2	-	-	-	-
DEKR [5]	single-stage	67.3	61.5	76.1	-	-	-	-	-	-	-
CID [20]	-	68.9	63.2	77.7	61.7	52.8	72.2	63.9	73.5	59.4	66.9

**Figure 2: The visualization results of HumVis.**

contains more contextual cues than features generated by hard cropping. It is hence used to infer instance-level cues for a specific person. Compared with previous pipelines, this new pipeline is robust to spatial detection errors, and is able to explore contextual cues at larger scales, thus achieving superior performance in human-centric visual analysis tasks.

Based on our CID [20], we build the human-centered visual analysis system named HumVis[†], which effectively separate different persons in RGB images to perform instance-level human analysis. As shown in Fig. 1, HumVis is a web-based system supporting two types of inputs, *e.g.*, live stream videos and uploaded videos. The live stream demo can read a video stream from camera and estimate corresponding multiple persons information in real-time. Users can also upload images or videos to get human pose and shape results.

2 THE HUMVIS SYSTEM

HumVis is a web-based demo system built with the CID and multiple task specific modules. The HumVis runs on a server equipped with RTX 3090 GPU, intel i7 CPU, and 128GB memory. As shown in Fig. 2, users can open a camera for live stream inputs, or upload new images and videos. HumVis provides interface to demonstrate the results of pose estimation, foreground segmentation, part segmentation and human mesh recovery, respectively. The following part briefly describes important algorithms implemented by HumVis.

2.1 Contextual Instance Decoupling

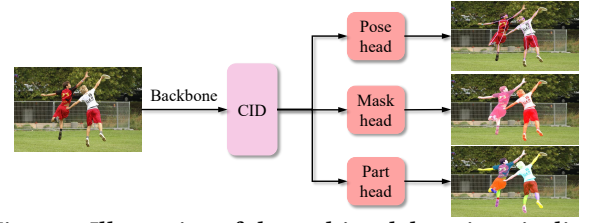
The core component is CID that separates persons into different feature maps for subsequent analysis tasks. Given an image with multiple persons, CID first extracts the location and feature for each person via Instance Information Abstraction (IIA), which are then used to decouple different persons in Global Feature Decoupling (GFD). GFD computes spatial and channel attention with location and feature cues of each person to isolate different persons. The generated instance-aware feature maps are capable of performing different tasks. More details about CID can be found from [20].

2.2 Multi-Task Learning Pipeline

In [20] we test CID with a separately trained model on each task, which requires large computation and memory cost. To improve the efficiency of HumVis, we propose a multi-task learning pipeline

Table 2: Efficiency comparison.

	Params(M)	GFLOPs	Speed(fps)	Pose(AP)
Single-Task	29.3	43.35	26.17	69.8
Separate	87.9	130.05	9.86	69.8
Multi-Task	29.4	45.75	24.27	69.5

**Figure 3: Illustration of the multi-task learning pipeline.**

that shares a large portion of model parameters for different tasks, and performs each task with a task-specific head. As shown in Fig. 3, we adopt the shared backbone and CID to extract instance-aware feature maps for each person and generate final results with different lightweight heads. Benefited by this multi-task learning pipeline, the inference efficiency is improved from 10 fps to 24 fps, while maintaining a similar performance, as shown in Table 2.

2.3 Applications

HumVis with CID can be applied into different human analysis tasks from crowded scenes. This part briefly introduces supported tasks by HumVis, and compares it with recent methods in Table 1.

Human Pose Estimation targets to detect all persons in an image and locates their keypoints. We estimate keypoint heatmap based on the generated instance-aware feature maps by CID.

Human Foreground Segmentation aims to generate the foreground segmentation mask for each person in image. We adopt Fully Convolutional Network (FCN) to generate mask by per-pixel classification with the instance-aware feature maps by CID.

Human Part Segmentation aims to parse person at fine-grained level, *i.e.*, generating part segmentation masks. We perform per-pixel multi-class classification on the instance-aware feature map from CID to generate part-level masks.

Body Mesh Recovery aims to reconstruct 3D human body mesh from inputs. We follow previous work [3] to directly regress the joint rotations and shape parameters based on the instance-aware features extracted by CID for each person.

3 CONCLUSION

This paper presents a human-centric visual analysis system named HumVis. HumVis is build on our CID and shows promising performance in multiple tasks including human pose estimation, and body and part segmentation, as well as 3D human mesh recovery.

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[†]Project page: <https://github.com/kennethwdk/HumVis>

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